

ROW-TOTAL CACHE: LOCAL FUNCTIONAL EVIDENCE

Candidate RTL SHA-256: 1e3fab94b7b2... | 26 RTL files | parent f028ec5

A Actual completion counts, with distinct units

Recorded tool	Loader transactions	Reducer input cases	Geometries: loader / reducer
Verilator 5.020	16,150	1,430,848	5 / 7
Verilator 5.49 dev	16,150	1,430,848	5 / 7

Both tools replay the same defined cases; counts are not added into one coverage total.

B Exact logged row totals and independently recomputed errors

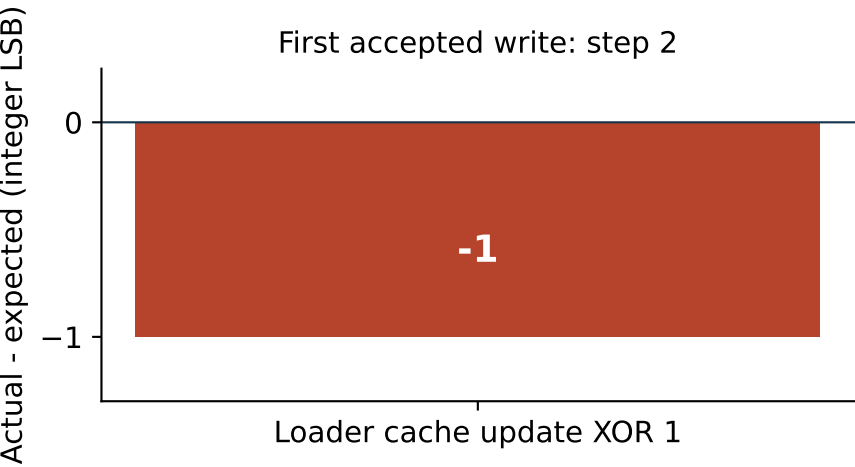
Slices x units	Maximum: row_last (hex)	After replacement: row0 (hex)	Integer errors*
1 x 1	00007fffffffffffff	00000000000000001	0 / 0
32 x 1	00007fffffffffffff	00000000000000001	0 / 0
18 x 71	00237ffffffffffffb9	0022ffffffffffffbb	0 / 0
1 x 128	003fffffffffffff80	003f7fffffffffff82	0 / 0
32 x 128	003fffffffffffff80	003f7fffffffffff82	0 / 0

* Actual - independent expected, maximum / replacement; both tools agree at all 10 logged points.

Expected maximum = $U (2^{47} - 1)$; after clear + replacement = $(U - 1)(2^{47} - 1) + 1$.

Values are exact stored integers. Only these two stages were logged; no clear-edge waveform is inferred.

C Two separate mutations are detected by independent checks



Consumer input: active row 0 XOR 1

Raw weights and scalar oracle remain unchanged.
Only the cached-row input is perturbed.

Detected at check 1:
cached DUT scalar mismatch

Both mutants return nonzero (recorded exit 1). Expected failures are not counted as passing samples.

Source: docs/evidence/20260930/row_cache/ | SHA index, original logs, exact integer CSV and source identity checked.

Local zero-delay functional evidence only. No new CI, mapped-netlist, FPGA/ASIC timing or PPA benefit is claimed.